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Started on Tuesday, 1 March 2022, 9:26 AM

State Finished

Completed on Tuesday, 1 March 2022, 10:22 AM

Time taken 56 mins 32 secs

Marks 20.50/42.00

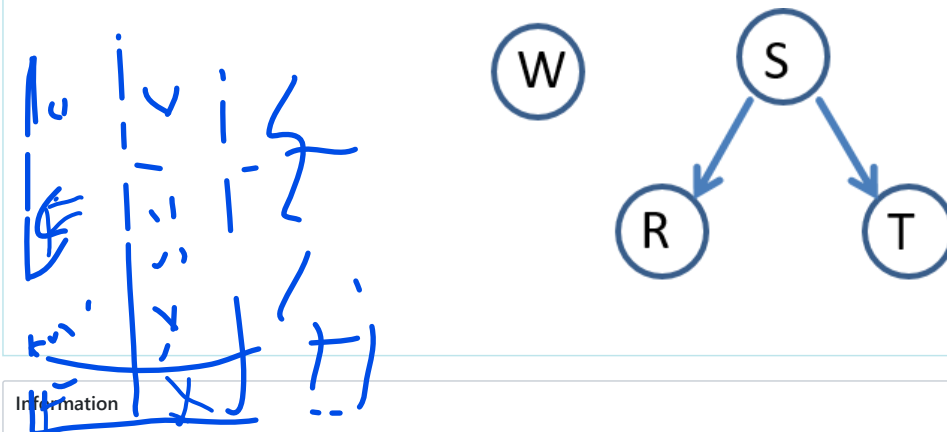
Grade 48.81 out of 100.00

Information

Consider a world defined by the following set $\mathcal{X}$ of random variables:

$$\begin{aligned} \text{Rain}(R) &= \{\text{yes}, \text{no}\} \\ \text{Temperature}(T) &= \{\text{low}, \text{high}\} \\ \text{Weekend}(W) &= \{\text{yes}, \text{no}\} \\ \text{Season}(S) &= \{\text{spring}, \text{summer}, \text{fall}, \text{winter}\} \end{aligned}$$

and (in)dependencies encoded by this Bayesian Network structure graph:



Information

If we exploit all the independencies implied by this network, how many *independent (non-redundant)* parameters (numbers) are strictly needed to specify ...

Question 1

Incorrect

Mark 0.00 out of 1.00

 $P(W, S)$

Answer:

$P(W, S) = P(W)P(S)$
 params for $P(W)$ $2^2 - 1 = 3$
 params for $P(S) = 1$
 $3 + 1 = 4$

$$\begin{array}{c|c} & -w \\ \hline * & \end{array}$$

$$\begin{array}{c|c|c|c} & S & & \\ \hline S & w & f & s \\ \hline + & + & + & \end{array}$$

The correct answer is:

Question 2

Incorrect

Mark 0.00 out of 1.00

 $P(S, R, T)$

Answer:

✗

The correct answer is:

Question 3

Incorrect

Mark 0.00 out of 1.00

 $P(W, S, R, T)$

Answer:

✗

params of $P(W) = 1$ 1 + params of $P(S, R, T)$

The correct answer is:

Information

Which of these statements about this model are correct? (Again: the inference questions below assume that we exploit all the independencies implied by the model, and don't blindly run the "standard" Inference by Enumeration algorithm.)

Question 4

Correct

Mark 1.00 out of 1.00

When calculating $P(T)$ via Inference by Enumeration, W can be completely ignored.

Select one:



The correct answer is

Question 5

Correct

Mark 1.00 out of 1.00

In any topological variable ordering for this network, W must come before T .

Select one:



The correct answer is

Question 6

Correct

Mark 1.00 out of 1.00

S, T, R, W is a valid topological variable ordering for this network.

Select one:



The correct answer is

Question 7

Correct

Mark 1.00 out of 1.00

This network encodes the joint distribution $P(W, S, R, T)$ as $P(W)P(R|S)P(T|S)$.

Select one:

The correct answer is **Question 8**

Incorrect

Mark 0.00 out of 1.00

If we added a link $T \rightarrow R$, the resulting network would not have a valid topological ordering any more.

Select one:

The correct answer is **Question 9**

Incorrect

Mark 0.00 out of 1.00

$P(T) = P(T|W)$

Select one:

The correct answer is 

Question 10

Correct

Mark 1.00 out of 1.00

Calculating $P(T|W)$ will require causal reasoning.

Select one:



The correct answer is

Question 11

Correct

Mark 1.00 out of 1.00

When calculating $P(R,T)$ via Inference by Enumeration, we only need to sum over S .

Select one:



The correct answer is

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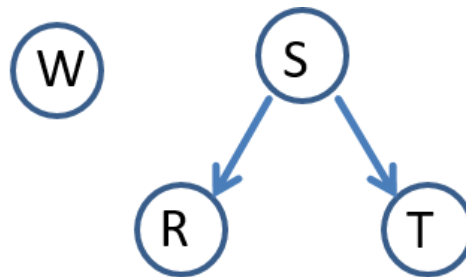
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Information

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and a Bayesian Network with this structure graph:



Information

Which of the following statements are correct?

Question 12

Correct

Mark 1.00 out of 1.00

In Likelihood Weighting for a query $P(W|R)$, the value sampled for the variable W will not have any effect on the final weight of the sampled event.

Select one:



The correct answer is ☐

Question 13

Incorrect

Mark 0.00 out of 1.00

In answering the query $P(S = \text{winter} \mid W)$ via exact inference, the result will be the same for all values of W .

Select one:



The correct answer is

Question 14

Correct

Mark 1.00 out of 1.00

In Likelihood Weighting for a query $P(S \mid R)$, the value sampled for the variable S will not have any effect on the final weight of the sampled event.

Select one:



The correct answer is

Question 15

Correct

Mark 1.00 out of 1.00

Given the query $P(R \mid W = \text{yes})$, Rejection Sampling will throw away all samples with values $W \neq \text{yes}$.

Select one:



The correct answer is

Question 16

Not answered

Marked out of 1.00

In Likelihood Weighting for a query $P(T|R)$, the value sampled for the variable S will not have any effect on the final weight of the sampled event.

Select one:



The correct answer is



Question 17

Correct

Mark 1.00 out of 1.00

Consider the query $\backslash(P(R | S)\backslash)$. Which of the following expression(s) correctly represent the Resampling Distribution for variable $\backslash(R)\backslash$?

☒ $\backslash(1/Z * P(S|R)P(S|T)\backslash)$
☒ $\backslash(1/Z * P(R|S)\backslash)$
☒ $\backslash(1/Z * P(S|W)P(R|S)P(T|S)\backslash)$
☒ $\backslash(1/Z * P(R|S)P(T|S)\backslash)$
☒ $\backslash(1/Z * P(S)P(R|S)P(T|S)\backslash)$
☒ $\backslash(1/Z * P(S|R)\backslash)$

Your answer is correct.

The correct answer is:


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Information

Consider the task of Bayesian Parameter Estimation for a model with one single *continuous* parameter $\theta \in [0, 1]$ and a dataset $\mathcal{D}$ over a set of *discrete* random variables $\mathcal{X}$. Which of the following statements are true?

Question 18

Incorrect

Mark 0.00 out of 1.00

With a non-uniform (= non-constant) prior, the MAP estimate cannot be the same as the ML estimate.

Select one:



The correct answer is

Question 19

Incorrect

Mark 0.00 out of 1.00

If we created a new dataset $\mathcal{D}2$ from $\mathcal{D}$ by duplicating each individual example in $\mathcal{D}$ (so that every example from $\mathcal{D}$ appears twice in $\mathcal{D}2$), the MAP estimates $\arg \max_{\theta} p(\theta|\mathcal{D})$ and $\arg \max_{\theta} p(\theta|\mathcal{D}2)$ will be the same.

Select one:



The correct answer is

Question 20

Correct

Mark 1.00 out of 1.00

If the prior $p(\theta)$ assigns a value of 1 to some specific value x of θ , then that x will obtain the highest posterior probability, regardless of the data.

Select one:



The correct answer is

Question 21

Incorrect

Mark 0.00 out of 1.00

$p(\mathcal{D})$ is irrelevant for finding the posterior mean estimate for θ .

Select one:



The correct answer is

Question 22

Incorrect

Mark 0.00 out of 1.00

If we use a Gaussian prior with a mean exactly in the middle of the parameter's range (at 0.5), then the Bayesian MAP estimate for θ will always give the same value as the Maximum Likelihood (ML) estimate.

Select one:



The correct answer is

Question **23**

Correct

Mark 1.00 out of 1.00

No dataset $\mathcal{D}$ can have a likelihood of strictly zero.

Select one:

The correct answer is [◀ PM 2021 Exchange](#)[Jump to...](#)[Exam Procedure and Rules ▶](#)

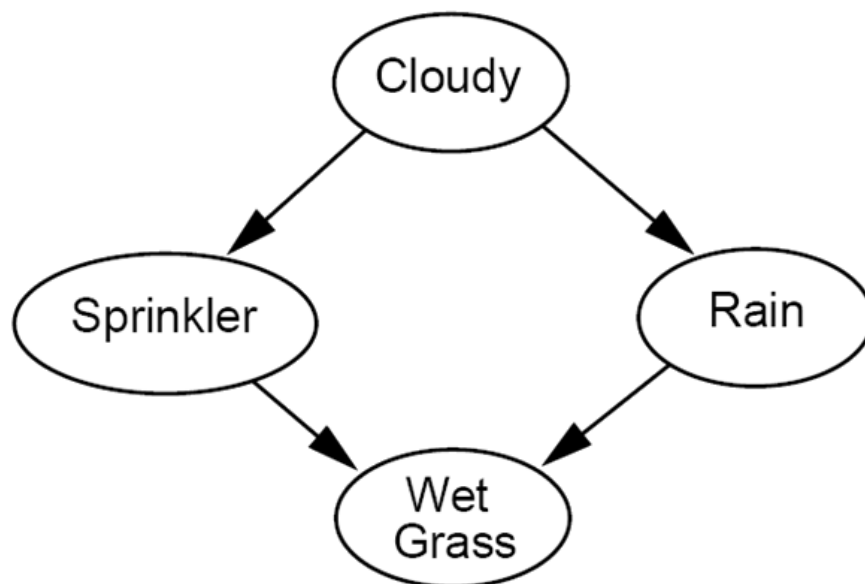
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Information

Consider the task of building a model that supports reasoning about the lawn in front of your house. The situation is described via four Boolean variables *Cloudy* (*C*), *Sprinkler* (*S*), *Rain* (*R*), and *WetGrass* (*W*).

Suppose we are in the process of learning a Bayesian network from a database $\mathcal{D}$ of example observations from this world. We are currently considering the following network structure, which has a certain likelihood $\mathcal{X}$.

(Note: When we speak of the likelihood of a model structure, we always mean the likelihood *given its corresponding ML parameters*):



Information

Suppose we add a link $S \rightarrow R$ to the above network. Which of the following statements are true, which are false?

Question 24

Incorrect

Mark 0.00 out of 1.00

This model could not be learned by any of the structure learning methods discussed in class.

Select one:



The correct answer is

Question 25

Correct

Mark 1.00 out of 1.00

The resulting network cannot have a lower likelihood on $\mathcal{D}$ than the original one.

Select one:



The correct answer is

Question 26

Incorrect

Mark 0.00 out of 1.00

The BIC Score for the resulting network structure cannot be lower than the BIC Score for the original structure.

Select one:



The correct answer is

Information

Suppose we believe there is a specific independency in the world (the sprinkler has no weather sensor and starts randomly) and force a structure learning algorithm to always leave C and S unconnected. Compared to a learner that is not limited in this way, does this constraint ...

Question 27

Incorrect

Mark 0.00 out of 1.00

improve the possibility for the learner to fit the training set?

Select one:



The correct answer is 

Question 28

Correct

Mark 1.00 out of 1.00

increase the bias of the learner?

Select one:



The correct answer is 

Question 29

Correct

Mark 1.00 out of 1.00

decrease the variance of the learner?

Select one:



The correct answer is 

Question **30**

Incorrect

Mark 0.00 out of 1.00

make it impossible for the learner to model correlations between S and C that may appear in the data?

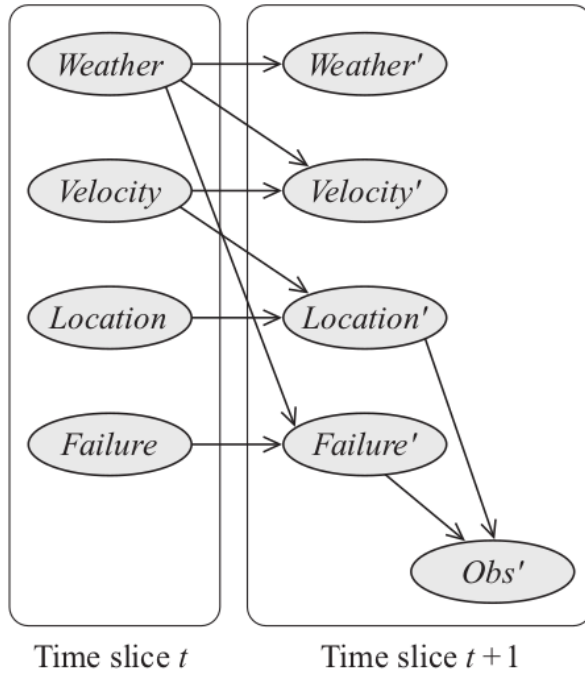
Select one:

The correct answer is [◀ PM 2021 Exchange](#)[Jump to...](#)[Exam Procedure and Rules ▶](#)

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Information

Consider once again the hypothetical Vehicle Localisation Network from the lecture slides:



Which of the following statements about this model are correct?

Question 31

Correct

Mark 1.00 out of 1.00

This is not a state-observation model because it violates the assumption of independence of observations from the previous state.

Select one:



The correct answer is [X]

Question 32

Correct

Mark 1.00 out of 1.00

$$P(Obs') = P(Obs' \mid Weather')$$

Select one:



The correct answer is

Question 33

Correct

Mark 1.00 out of 1.00

This is not a valid state-observation model because state variable V depends on more than two previous state variables.

Select one:



The correct answer is

Question 34

Correct

Mark 1.00 out of 1.00

The motivation for including the arrow from *Failure* to *Failure'* is to model correlations of failure states over time.

Select one:



The correct answer is

Information

Suppose we modeled the (1D) vehicle motion as a Kalman Filter, with the (continuous) hidden state variables *Weather*, *Velocity*, *Location*, *Failure*, and the (continuous) observation variable *Obs*.

[Note: Assume there is some meaningful way of encoding *Weather* and *Failure* as numbers.]

Question 35

Incorrect

Mark 0.00 out of 1.00

What is the size (# of rows/columns) of the state transition matrix **A**?

Number of rows:

✖

Number of columns:

✖

Question 36

Partially correct

Mark 0.50 out of 1.00

What is the size (# of rows/columns) of the observation model matrix **B**?

Number of rows:

✔

Number of columns:

✖

Question 37

Incorrect

Mark 0.00 out of 1.00

How many zeroes (0) does the observation model matrix **B** contain (at least)?

Answer:



✖

The correct answer is:



Question 38

Incorrect

Mark 0.00 out of 1.00

What is the maximum number of maxima that the forward message $p(\mathbf{x}^{(t)} | \mathbf{z}^{(1:t)})$ can have at any point t ?

0
1
2
3
4
5
8
9
 $\lim_{t \rightarrow +\infty} 4^t$
+∞

Your answer is incorrect.

The correct answer is:

Information

Now let's go back to our even simpler bicycle model from the lecture slides:



$$\begin{aligned} p^{(t+1)} &= p^{(t)} + v^{(t)} + \epsilon_p \\ v^{(t+1)} &= v^{(t)} + \epsilon_v \\ z^{(t)} &= p^{(t)} + \epsilon_z \end{aligned}$$

$$\text{with } \epsilon_p \sim \mathcal{N}(0; \sigma_p^2); \quad \epsilon_v \sim \mathcal{N}(0; \sigma_v^2); \quad \epsilon_z \sim \mathcal{N}(0; \sigma_z^2)$$

Assume we have an observation sequence $\mathbf{z} = \{z, z, z, \dots\}$, i.e., a sequence (of arbitrary length) of identical values. Which of the following statements are true:

Question 39

Incorrect

Mark 0.00 out of 1.00

The forward message will increase the value of ϵ_v with each iteration.

Select one:



The correct answer is

Question 40

Incorrect

Mark 0.00 out of 1.00

The mean of $p(V^{(t)})$ will converge to z , for growing t .

Select one:



The correct answer is

Question 41

Incorrect

Mark 0.00 out of 1.00

The forward message will converge to a normal distribution with mean vector $\mu^{(t)} = (z, 0)^T$, for growing t .

Select one:



The correct answer is

Question **42**

Correct

Mark 1.00 out of 1.00

The forward message will become flatter (broader) with each time step.

Select one:



The correct answer is

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